

External Consistency. Evaluates surface-level coherence using **External Consistency Verification** ($\mathbb{E}[\mathbb{I}(\text{no violation})]$), a LLM-as-a-Judge approach following [1].

3.4 Training Framework

For the number guessing task, we employ supervised fine-tuning using data from our prompt-based experiments. We curate the training set by selecting dialogues that exhibit no external inconsistencies, ensuring that the models we train at least maintain surface-level coherence. The training objective combines KL divergence and cross-entropy losses: $\mathcal{L} = \sum_{t=1}^T [\alpha \cdot D_{\text{KL}}(P_{\text{probe}}^t \| P_{\text{initial}}) + \beta \cdot \mathcal{L}_{\text{CE}}]$, where P_{initial} is the belief distribution from the first probe, P_{probe}^t is the distribution at turn t , and $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss for the probing responses. We also experiment with each loss component in isolation ($\alpha = 0$ or $\beta = 0$) to analyze their individual effects on consistency improvement.

4 Experimental Results

4.1 Prompting

Table 1: Prompting results. "+ Reasoning" stands for using the reasoning model as the Proposer.

Model	Number Guessing				Entity Guessing			
	D.R. (↓)	Once D.R. (↓)	KL Div. (↓)	E.C.V. (↓)	D.R. (↓)	Once D.R. (↓)	KL Div. (↓)	E.C.V. (↓)
GPT-4o	43.36 $\pm$ 11.72	100.00 $\pm$ 0.00	0.66 $\pm$ 0.27	0.00 $\pm$ 0.00	22.64 $\pm$ 8.46	100.00 $\pm$ 0.00	0.75 $\pm$ 0.43	0.00 $\pm$ 0.00
Seed-1.6	17.37 $\pm$ 5.62	95.24 $\pm$ 21.30	0.99 $\pm$ 0.38	42.86 $\pm$ 49.49	11.31 $\pm$ 4.52	100.00 $\pm$ 0.00	0.17 $\pm$ 0.05	0.00 $\pm$ 0.00
+ Reasoning	28.57 $\pm$ 7.49	94.12 $\pm$ 23.53	-	0.00 $\pm$ 0.00	11.05 $\pm$ 5.18	95.12 $\pm$ 21.54	-	4.88 $\pm$ 21.54
Deepseek-v3.1	38.46 $\pm$ 13.71	100.00 $\pm$ 0.00	0.62 $\pm$ 0.23	80.95 $\pm$ 39.27	37.68 $\pm$ 11.38	100.00 $\pm$ 0.00	0.09 $\pm$ 0.04	41.46 $\pm$ 49.27
+ Reasoning	54.25 $\pm$ 21.56	95.24 $\pm$ 21.30	-	28.57 $\pm$ 45.18	27.01 $\pm$ 13.54	100.00 $\pm$ 0.00	-	7.32 $\pm$ 26.04
Claude-3.7-Sonnet	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	-	0.00 $\pm$ 0.00	12.25 $\pm$ 2.11	100.00 $\pm$ 0.00	-	2.86 $\pm$ 16.66
+ Reasoning	71.15 $\pm$ 18.30	100.00 $\pm$ 0.00	-	0.00 $\pm$ 0.00	33.42 $\pm$ 12.61	100.00 $\pm$ 0.00	-	2.44 $\pm$ 15.43

Our experiments reveal pervasive implicit inconsistency across all tested LLMs (Table 1; see Appendix D for details). Most models exhibit 100% Once Drift Rate (See Appendix C), with Drift Rates ranging from 17.37% to 100%. We hypothesize that reasoning-enhanced models increase drift in simple tasks (like Number Guessing) due to overthinking—introducing unnecessary complexity in straightforward scenarios. Claude’s reversed pattern may suggest architecture-specific differences in how reasoning modules interact with goal maintenance; we keep it as our future work. Notably, reasoning improves external consistency in some cases (e.g., Deepseek-v3.1 reduces violations), indicating that while reasoning may destabilize implicit goals, it can enhance surface-level coherence.

4.2 Supervised Fine Tuning

Due to computational constraints, we fine-tune Qwen-2.5-14B-Instruct as Proposer with Seed-1.6-Reasoning as Guesser. Results (See Table 2) show CE-only fine-tuning fails to mitigate implicit inconsistency, while KL-only significantly improves goal persistence. The combined CE+KL approach yields the compromise results, indicating KL divergence effectively regularizes belief stability.

Table 2: Qwen-2.5-14B-Instruct variants metric results in Number Guessing.

Model	D.R. (↓)	O.D.R. (↓)	KL Div. (↓)	E.C.V. (↓)
Qwen-2.5-14B-Instruct	36.83 $\pm$ 10.72	100.00 $\pm$ 0.00	3.30 $\pm$ 3.15	100.00 $\pm$ 0.00
+ CE	31.63 $\pm$ 18.78	76.19 $\pm$ 42.59	0.47 $\pm$ 0.26	14.29 $\pm$ 34.99
+ KL	13.99 $\pm$ 9.04	95.24 $\pm$ 21.30	1.41 $\pm$ 1.11	100.00 $\pm$ 0.00
+ CE & KL	14.48 $\pm$ 5.08	95.24 $\pm$ 21.30	1.46 $\pm$ 0.97	95.24 $\pm$ 21.30

5 Conclusion and Future Work

In this work, we identified and formalized a critical limitation in current LLMs: the inability to maintain implicit consistency—stable adherence to unstated goals across multi-turn interactions. Through a carefully designed 20-question paradigm and unified probing framework, we demonstrated

that state-of-the-art models universally exhibit goal drift, changing their internal targets in a significant portion of dialogue turns.

Our findings reveal that external behavioral consistency does not guarantee internal goal stability, highlighting a fundamental challenge for developing truly reliable persona-driven AI. While we showed that explicit training with KL divergence regularization can mitigate this issue, the pervasive nature of implicit inconsistency suggests that more sophisticated architectural solutions are needed.

Future work should explore memory mechanisms, explicit belief state tracking, and alternative training objectives to anchor implicit goals over time. As LLMs increasingly power interactive applications, ensuring their internal consistency becomes not just a technical challenge, but a prerequisite for trustworthy human-AI interaction.

References

- [1] Kartikeya Badola, Jonathan Simon, Arian Hosseini, Sara Marie Mc Carthy, Tsendsuren Munkhdalai, Abhimanyu Goyal, Tomáš Kočiský, Shyam Upadhyay, Bahare Fatemi, and Mehran Kazemi. Multi-turn puzzles: Evaluating interactive reasoning and strategic dialogue in llms, 2025. URL <https://arxiv.org/abs/2508.10142>.
- [2] Harsh Raj, Vipul Gupta, Domenic Rosati, and Subhabrata Majumdar. Improving consistency in large language models through chain of guidance, 2025. URL <https://arxiv.org/abs/2502.15924>.
- [3] Jingyuan Yang, Dapeng Chen, Yajing Sun, Rongjun Li, Zhiyong Feng, and Wei Peng. Enhancing semantic consistency of large language models through model editing: An interpretability-oriented approach. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3343–3353, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.199. URL <https://aclanthology.org/2024.findings-acl.199/>.
- [4] Wikipedia. Twenty questions — Wikipedia, the free encyclopedia. <http://en.wikipedia.org/w/index.php?title=Twenty%20questions&oldid=1312122168>, 2025. [Online; accessed 02-November-2025].
- [5] Zhenru Lin, Jiawen Tao, Yang Yuan, and Andrew Chi-Chih Yao. Existing llms are not self-consistent for simple tasks, 2025. URL <https://arxiv.org/abs/2506.18781>.
- [6] Marcus J. Min, Yangruibo Ding, Luca Buratti, Saurabh Pujar, Gail Kaiser, Suman Jana, and Baishakhi Ray. Beyond accuracy: Evaluating self-consistency of code large language models with identitychain. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *International Conference on Representation Learning*, volume 2024, pages 5454–5469, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/file/16371a9d5fed65d6d78ca3a7fa6e598c-Paper-Conference.pdf.
- [7] Yizhe Zhang, Jiarui Lu, and Navdeep Jaitly. Probing the multi-turn planning capabilities of LLMs via 20 question games. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1495–1516, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.82. URL <https://aclanthology.org/2024.acl-long.82/>.
- [8] Yuyan Chen, Yuezhe Li, Songzhou Yan, Sijia Liu, Jiaqing Liang, and Yanghua Xiao. Do large language models have problem-solving capability under incomplete information scenarios? In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2225–2238, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.131. URL <https://aclanthology.org/2024.findings-acl.131/>.
- [9] Qi Chen, Bowen Zhang, Gang Wang, and Qi Wu. Weak-eval-strong: Evaluating and eliciting lateral thinking of llms with situation puzzles, 2024. URL <https://arxiv.org/abs/2410.06733>.

- [10] Davide Mazzaccara, Alberto Testoni, and Raffaella Bernardi. Learning to ask informative questions: Enhancing LLMs with preference optimization and expected information gain. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 5064–5074, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.291. URL <https://aclanthology.org/2024.findings-emnlp.291/>.
- [11] Alvin Dey, Harsh Kumar Jain, Vikash Kumar Pandey, and Tanmoy Chakraborty. All it takes is 20 questions!: A knowledge graph based approach, 2019. URL <https://arxiv.org/abs/1911.05161>.
- [12] Huang Hu, Xianchao Wu, Bingfeng Luo, Chongyang Tao, Can Xu, Wei Wu, and Zhan Chen. Playing 20 question game with policy-based reinforcement learning. In Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun’ichi Tsujii, editors, *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 3233–3242, Brussels, Belgium, October–November 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1361. URL <https://aclanthology.org/D18-1361/>.
- [13] Junhyuk Choi, Yeseon Hong, Minju Kim, and Bugeun Kim. Examining identity drift in conversations of llm agents, 2025. URL <https://arxiv.org/abs/2412.00804>.
- [14] Rauno Arike, Elizabeth Donoway, Henning Bartsch, and Marius Hobbhahn. Technical report: Evaluating goal drift in language model agents, 2025. URL <https://arxiv.org/abs/2505.02709>.
- [15] Michal Kosinski. Evaluating large language models in theory of mind tasks, 2024. URL <https://arxiv.org/abs/2302.02083>.

A External Inconsistency and Implicit Inconsistency

Figure 1 illustrates 2 distinct types of inconsistencies we identify: **External / Implicit Inconsistency**.

External Inconsistency (Left Panel) The model initially selects "Arctic Bear" as its target. Despite providing consistent answers to previous questions (e.g., answering "No" to "Does it have any white parts (excluding eyes)?"), it contradicts itself when directly asked "Is it Arctic Bear?" This creates an external logical inconsistency within the dialogue.

Implicit Inconsistency (Right Panel) The model’s internal belief shifts from "Arctic Bear" to "Panda" during the dialogue. Although all its answers remain factually correct for both entities, the change in its internal target constitutes an implicit inconsistency, as the model fails to maintain its original goal.

B Related Works

LLM Consistency Evaluation Existing research on LLM consistency has predominantly focused on *external* aspects, such as factual coherence in multi-turn dialogues [1–3, 5]. Although some work begins to address internal consistency, notably [6] examining self-consistency in code LLMs across tasks, these approaches focus on dual task alignment rather than the temporal stability of implicit goals within dialogues, which remains an unexplored dimension of model behavior.

20-Question-Style Problem for LLMs The 20-question paradigm has been widely used to evaluate LLMs’ reasoning and planning capabilities, focusing on how models generate strategic questions to narrow down the answer space [7–12]. However, these studies primarily assess the external behavior of planning. Our work repurposes this paradigm to probe the model’s internal state, specifically the stability of its implicitly selected goal throughout the dialogue.

Cognitive Foundations of Goal Persistence From a cognitive science perspective, *goal persistence* is a core component of executive function and theory of mind [13–15]. Human cognition maintains implicit objectives enabling consistent behavior over time, our work examining whether models exhibit similar stability in maintaining simple, self-selected goals during multi-turn interactions.